achievable operating trim or heel angle, has a rigid hull structure which penetrates the surface of the sea over more than one discrete area .
. 4 "Permeability" of a space means the percentage of the volume of that space which can be occupied by water .
. 5 "Skirt" means a downwardly-extending, flexible structure used to contain or divide an air cushion .
. 6 "Watertight" in relation to a structure means capable of preventing the passage of water through the structure in any direction under the head of water likely to occur in the intact or damaged condition .
7 "Weathertight" means that water will not penetrate into the craft in any wind and wave conditions up to those specified as critical design conditions.
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critical design conditions.
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2.2 Intact buoyancy
2.2.1 All craft should have a sufficient reserve of buoyancy at the design waterline to meet the intact and damage stability requirements of this chapter. The Administration may require a larger reserve of buoyancy to permit the craft to operate in any of its intended modes. This reserve of buoyancy should be calculated by including only those compartments which are:
2.2.2
2.2.3
2.2.4
.1 watertight;
.2 accepted as having scantlings and arrangements adequate to maintain their watertight integrity; and
.3 situated in locations below a datum, which may be a watertight deck or equivalent structure of a non-watertight deck covered by a weathertight structure as defined in 2.2.3.1 .
Arrangements should be provided for checking the watertight integrity of those compartments taken into account in 2.2.1.
Where entry of water into structures above the datum as defined in 2.2.1.3 would significantly influence the stability and buoyancy of

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integrity of those compartments taken into account in 2.2.1.
Where entry of water into structures above the datum as defined in 2.2.1.3 would significantly influence the stability and buoyancy of the craft, such structures should be:
. 1 of adequate strength to maintain the weathertight integrity and fitted with weathertight closing appliances; or
.2 provided with adequate drainage arrangements; or
.3 an equivalent combination of both measures.
The means of closing openings in the boundaries of weathertight structures should be such as to maintain weathertight integrity in all operational conditions.
2.3 Intact stability in the displacement mode
2.3.1 Hydrofoil craft fitted with surface-piercing foils and/or fully submerged foils should have sufficient stability under all permitted cases of loading to comply with the relevant provisions of annex 6 and specifically maintain a heel angle of less than 10 0 when subjected to the greater of the heeling moments in 1.1.2 and 1.1.4 of that annex.
2.3.2 Multihull craft should meet the relevant requirements of annex 7 for all permitted cases of loading.

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subjected to the greater of the heeling moments in 1.1.2 and 1.1.4 of that annex.
2.3.2 Multihull craft should meet the relevant requirements of annex 7 for all permitted cases of loading.
2.3.3 Subject to 2.3.4, all other craft should meet the following criteria in all permitted conditions of loading:
.1 resolution A.562(14) (weather criterion);

.2 the area under the righting lever curve (GZ curve) should not be less than 0.07 m.rad up to e = 15° when the maximum righting lever (GZ) occurs at e = 15° and 0.055 m.rad up to e = 30° when the maximum righting lever occurs at e = 30° or above. Where the maximum righting lever occurs at angles of between e = 15° and e = 30°, the corresponding area under the righting lever curve should be:
A = 0.055 + 0.001 (30° - emax) (m.rad)
where:
emax is the angle of heel in degrees at which the righting lever curve reaches its maximum;
.3 the area under the righting lever curve between e = 30° and e = 40° or between e = 30° and the angle of flooding

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where:
emax is the angle of heel in degrees at which the righting
lever curve reaches its maximum;
.3 the area under the righting lever curve between e = 30° and
e = 40° or between e = 30° and the angle of flooding
ef *, if this angle is less than 40°, should not be less
than 0.03 m.rad;
.4 the righting lever GZ should be at least 0.20 m at an angle of
heel equal to or greater than 30°;
.5 the maximum righting lever should occur at an angle of heel
not less than 15°; and
.6 the initial metacentric height GMo should not be less
than 0.15 m.
2.3.4 Where the characteristics of the craft are unsuitable for
application of 2.3.3, the Administration may accept alternative
criteria equivalent to those stipulated in 2.3.3, appropriate to
the type of craft and area of operation.
2.4 Intact stability in the non - displacement mode
2.4.1 The requirements of this section and section 2.12 should be applied
on the assumption that any stabilization systems fitted are fully
operational.
2.4.2 Suitable calculations should be carried out and/or tests conducted
to demonstrate that, when operating in the non-displacement and

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on the assumption that any stabilization systems fitted are fully
operational.
2.4.2 Suitable calculations should be carried out and/or tests conducted
to demonstrate that, when operating in the non-displacement and
transient modes within approved operational limitations, the craft
will, after a disturbance causing roll, pitch, heave or heel due to
turning or any combination thereof, return to the original attitude.
2.4.3 The roll and pitch stability on the first and/or any other craft of
a series should be qualitatively assessed during operational safety
trials as required by chapter 18 and annex 8. The results of such
trials may indicate the need to impose operational limitations.
In applying this criterion, small openings through which progressive
flooding cannot take place need not be considered as open.
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2.4.4 Where craft are fitted with surface piercing structure or
appendages, precautions should be taken against dangerous attitudes

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2.4.4 Where craft are fitted with surface piercing structure or appendages, precautions should be taken against dangerous attitudes or inclinations and loss of stability subsequent to a collision with a submerged or floating object.
2.4.5 In designs where periodic use of cushion deformation is employed as a means of assisting craft control, or periodic use of cushion air exhausting to atmosphere for purposes of craft manoeuvring, the effects upon cushion-borne stability should be determined, and the limitations on the use by virtue of craft speed or attitude should be established.
2.4.6 In the case of an air-cushion vehicle fitted with flexible skirts, it should be demonstrated that the skirts remain stable under operational conditions.
2.5 Intact stability in the transient mode
2.5.1 Under weather conditions up to the worst intended conditions, the time to pass from the displacement mode to the non-displacement mode and vice versa should be minimized unless it is demonstrated that no